WEEK 1 INTERNSHIP TASK

Data Acquisition and Pre-processing in Python

Project: Customer Churn Data Pre-processing

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Domain: Data Analytics and Machine Learning
Week: 1

Internship Name :- **Virtual Data Science Apprentice Intern**

Introduction

Data preprocessing is an important stage in the data science and machine learning workflow. Raw datasets may contain missing values, inconsistent data types, duplicate records, categorical variables, and numerical variables with different scales. Proper preprocessing helps improve data quality and prepares the dataset for further analysis and machine learning.

The objective of this Week 1 task is to develop a complete data acquisition and preprocessing pipeline using Python. A customer churn dataset was selected because it contains demographic, service usage, contract, billing, and customer retention information that can be used for data analysis and machine learning.

Objective

1. Acquire a publicly available dataset.
2. Understand the structure and characteristics of the dataset.
3. Identify and handle missing values.
4. Check for duplicate records.
5. Convert inappropriate data types.
6. Detect potential outliers.
7. Encode categorical variables.
8. Normalize numerical features.
9. Analyze relationships between features and the target variable.
10. Validate the final processed dataset.
11. Document the complete preprocessing workflow.

Dataset selection

The dataset selected for this project is a **Customer Churn dataset** containing information about customers, their demographic characteristics, services, contracts, billing information, and churn status.

Attributes	Description
Dataset	Customer Churn Dataset
Records	7,043
Original Features	21
Target Variable	Churn
Datatype	Mixed Categorical and Numerical
Domain	Customer Analytics/Telecommunications

Customer churn prediction is an important business problem because retaining existing customers can be more cost-effective than acquiring new customers. The dataset provides information that can be used to understand customer behavior and identify factors associated with churn.

[Source]: "https://raw.githubusercontent.com/IBM/telco-customer-churn-on-icp4d/master/data/Telco-Customer-Churn.csv"

Tools and Technology Used

Tool/Library	Purpose
Python	Programming Language
Pandas	Data loading , cleaning and manipulation
Numpy	Numerical Operations
Matplotlib	Data visualizations
Scikit-learn	Scaling and Preprocessing
Google Colab	Development Environment

Data Acquisition

The dataset was obtained from a publicly available data repository. The data was downloaded in CSV format and loaded into Python using the Pandas library.

Code:

```
url = "https://raw.githubusercontent.com/IBM/telco-customer-churn-on-icp4d/master/data/Telco-Customer-Churn.csv"

df = pd.read_csv(url)


print("Dataset loaded successfully")

print("Shape:",df.shape)

df.head()
```

The dataset was successfully loaded into a Pandas DataFrame. The initial dataset contained 7,043 records and 21 columns.

Initial Data Exploration

1.df.shape

Shape: (7043, 21)

2. df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

3. df.head()

```
Dataset loaded successfully
Shape: (7043, 21)
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	TechSupport	StreamingTV
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	...	No	No	No
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	...	Yes	No	No
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	...	No	No	No
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	...	Yes	Yes	No
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	...	No	No	No

5 rows × 21 columns

Initial exploration was performed to understand the number of observations, variables, data types, and overall structure of the dataset.

Missing Values Analysis

During data cleaning, the TotalCharges column was converted from object/string format to numeric format. Invalid or blank values were converted to NaN using errors="coerce". This revealed 11 missing values. These records were then handled during the cleaning process, resulting in zero missing values in the cleaned dataset.

df.isnull().sum()

```
df.isnull().sum()
```

	0
customerID	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0
tenure	0
PhoneService	0
MultipleLines	0
InternetService	0
OnlineSecurity	0
OnlineBackup	0
DeviceProtection	0
TechSupport	0
StreamingTV	0
StreamingMovies	0
Contract	0
PaperlessBilling	0
PaymentMethod	0
MonthlyCharges	0
TotalCharges	0
Churn	0

dtype: int64

Code for conversion

```
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")

print(df["TotalCharges"].dtype)

print("Missing values after conversion:",
    df["TotalCharges"].isnull().sum())
```

output

float64

Missing values after conversion: 11

Duplicate Analysis

Duplicate records were checked using the complete original dataset, including the unique customerID field. No exact duplicate customer records were found.

Output

Duplicates in original dataset: 0

During feature transformation, the unique customerID column was removed because it is an identifier rather than a predictive feature. After removing this identifier, some customers had identical values across all remaining features. This resulted in 22 rows being identified as duplicate feature records.

Output

Duplicate feature profiles: 33

Profiles with different Churn values: 18

Further investigation showed 33 repeated feature profiles, with 18 profiles having different churn outcomes. This confirmed that these were not necessarily duplicate customer records. They represented different customers who happened to have identical observable characteristics. Therefore, these records were retained rather than deleted.

Datatype Conversion

```
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")
```

```
print(df["TotalCharges"].dtype)


print("Missing values after conversion:",
      df["TotalCharges"].isnull().sum())
```

Numerical variables were converted to appropriate numerical data types to enable statistical analysis and machine learning preprocessing.

Outlier Detection

Method : I have used the IQR method to detect the outliers

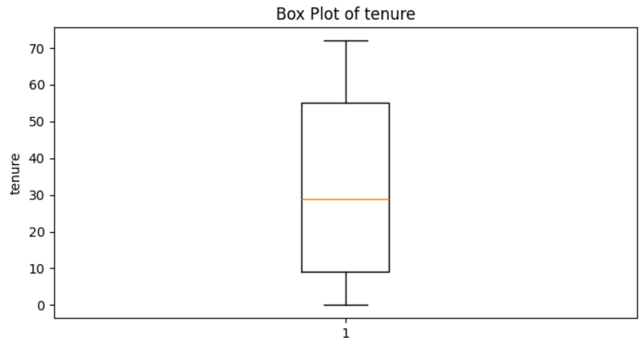
IQR = Q3-Q1

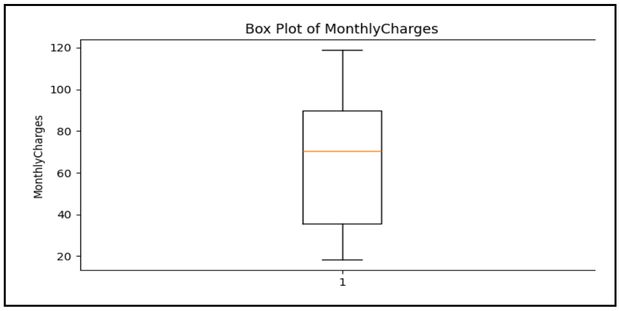
Lower Bound = Q1-1.5(IQR)

Upper Bound = Q3+1.5(IQR)

Feature	Number of outliers
tenure	0
MonthlyCharges	0
TotalCharges	0

The IQR method identified no outliers in tenure, MonthlyCharges, or TotalCharges. Box plots were also used to visually verify the results. Since no outliers were detected, no observations were removed or modified during this stage.





Categorical Encoding

Code:

```
binary_columns = [
    "gender",
    "Partner",
    "Dependents",
    "PhoneService",
    "PaperlessBilling",
    "Churn"
]
for column in binary_columns:
    df_cleaned[column] = df_cleaned[column].map({
        "Yes": 1,
        "No": 0,
        "Male": 1,
        "Female": 0
    })
```

One-hot coding

One-hot encoding was used for categorical variables containing multiple categories. The drop_first=True option was used to retain a reference category and reduce redundant encoded variables.

Identifier Removal

```
df_cleaned = df_cleaned.drop("customerID", axis=1)
```

customerID was removed because it uniquely identifies individual customers but does not provide meaningful predictive information for churn modeling. Retaining such identifiers could introduce unnecessary features without contributing useful behavioral information.

Feature Scaling/Normalization

Feature	Min	Max
tenure	0	72
MonthlyCharges	18.25	118.75

TotalCharges	0	8684.80

The numerical variables had substantially different ranges, particularly TotalCharges, which had a maximum value of 8684.80. To place the continuous numerical variables on a comparable scale, Min-Max normalization was applied.

Code:

```
from sklearn.preprocessing import MinMaxScaler


scaler = MinMaxScaler()


df_cleaned[scaling_columns] = scaler.fit_transform(
    df_cleaned[scaling_columns]
)
```

Result

Feature	Scaled Minimum	Scaled Maximum
tenure	0	1
MonthlyCharges	0	1
TotalCharges	0	1

After transformation, all three continuous numerical features were scaled to the 0–1 range.

Feature Correlation Analysis

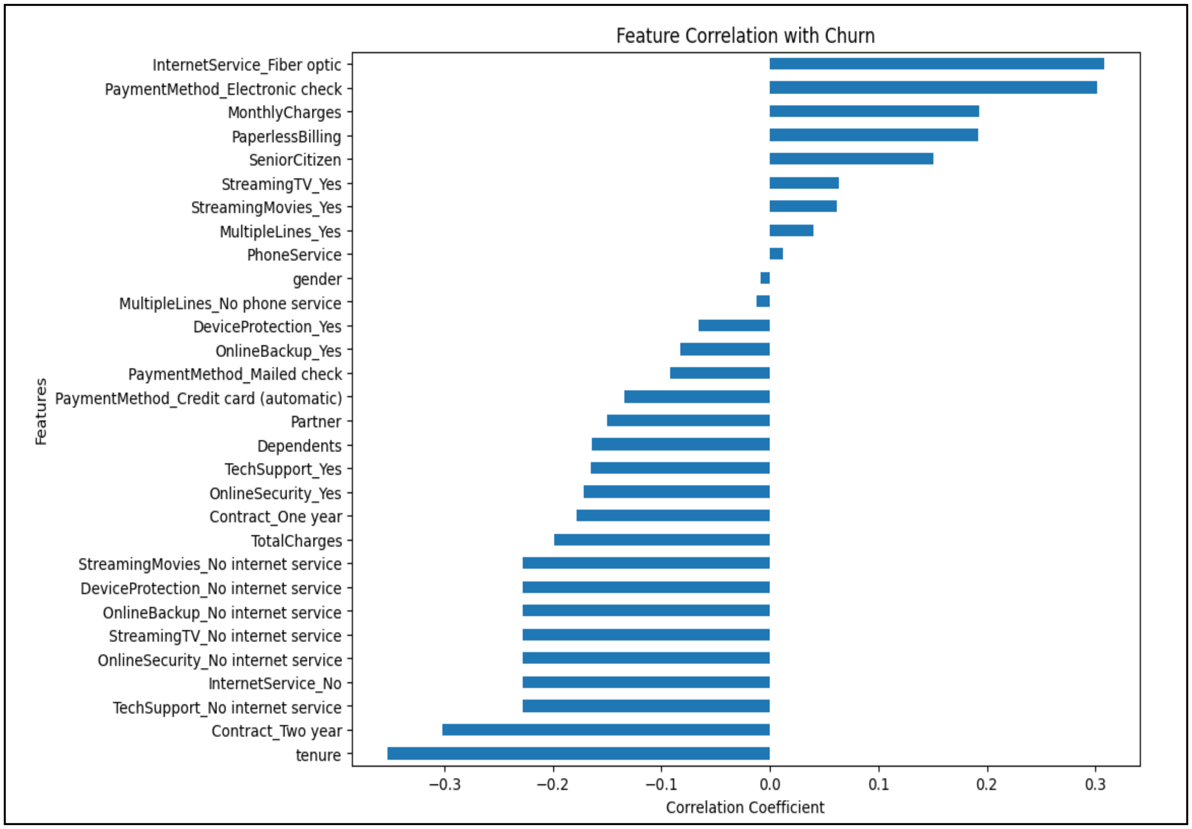
Positive Correlations

- InternetService_Fiber optic → 0.308
- Paymentmethod_Electronic check-> 0.302
- MonthlyCharges-> 0.193
- Paperlessbilling-> 0.192

Negative Correlations

- Tenure->-0.352
- Contract_Two_year->-0.302
- InternetService_No->-0.228

Pearson correlation analysis was performed to understand the relationship between the available features and the target variable, Churn. The strongest negative relationship was observed for tenure with a correlation coefficient of approximately −0.352. Contract_Two year also showed a relatively strong negative relationship. On the positive side, InternetService_Fiber optic and PaymentMethod_Electronic check showed relatively stronger positive relationships with churn.Correlation was used for exploratory feature analysis rather than automatic feature elimination. Features with weak individual correlations were retained because they may still provide useful information when combined with other variables in a machine learning model.



Final Dataset Validation

FINAL DATASET VALIDATION

Shape: (7043, 31)

Missing values: 0

Exact duplicate rows: 22

Object columns: 0

Infinite values: 0

Challenges Encountered and Solutions

Challenge 1 — TotalCharges stored as object

Problem:
TotalCharges was initially stored as an object/string column rather than a numerical column.

Solution:
Converted it using pd.to_numeric() and handled the resulting missing values.

Challenge 2 — Missing values

Problem:
Conversion of TotalCharges revealed 11 missing values.

Solution:
The missing values were handled during the cleaning stage, resulting in zero missing values.

Challenge 3 — Duplicate feature profiles

Problem:
After removing customerID, 22 rows appeared as duplicates.

Investigation:
33 repeated feature profiles were identified, and 18 had different churn outcomes.

Solution:
The rows were retained because they represented distinct customers with potentially identical characteristics.

Challenge 4 — Different numerical scales

Problem:
TotalCharges had a much larger numerical range than tenure and MonthlyCharges.

Solution:
Min-Max scaling was applied to the continuous numerical features.

Conclusion

The Week 1 task successfully established a complete data acquisition and preprocessing pipeline using Python. The customer churn dataset was inspected, cleaned, transformed, and validated using Pandas, NumPy, Matplotlib, and Scikit-learn.

Missing values were handled, data types were corrected, and duplicate records were carefully investigated. Outlier analysis using the IQR method identified no outliers in the selected numerical features. Categorical variables were converted into numerical representations using binary and one-hot encoding. Continuous numerical features were normalized using Min-Max scaling.

Correlation analysis provided additional insights into the relationship between customer characteristics and churn. The final dataset contained 7,043 records and 31 numerical columns, with no missing values, object-type columns, or infinite values.

The resulting dataset is therefore prepared for further exploratory analysis and machine learning modeling.

Future Scope

The preprocessed dataset can be used for further exploratory data analysis and machine learning tasks. Future work could include analyzing churn patterns across customer segments, performing feature importance analysis, splitting the dataset into training and testing sets, and developing classification models such as Logistic Regression, Decision Trees, Random Forest, or Gradient Boosting. Model evaluation can then be performed using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC.

Appendix

1. Data loading
2. Dataset inspection
3. Missing-value handling
4. Duplicate checking
5. Data type conversion
6. Outlier detection
7. Identifier removal

8. Binary encoding
9. One-hot encoding
10. Min-Max scaling
11. Correlation analysis
12. Final validation